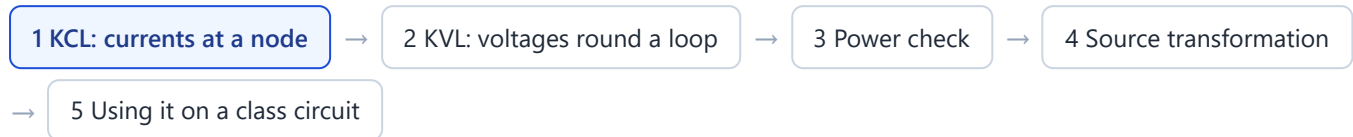


03 — KVL, KCL and source transformation

Class notes p.4–5, set 2 p.4, p.11 · MTE syllabus item 3 · 1 board question + 1 textbook problem · needs 02

Map



The questions this file answers

- **textbook** A loop with 12 V, 2 k Ω , an opposing 4 V source and 4 k Ω : find i , each voltage, and each element's power.
- **Class p.4** Source transformation: V in series with R $\Leftrightarrow$ V/R in parallel with R.
- **Class p.7 · set 2 p.8** The superposition circuit again, solved in three lines by source transformation.

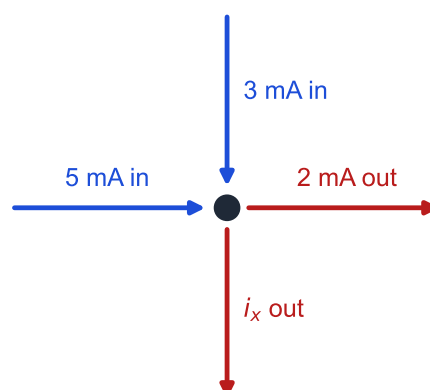
The professor used both laws inside other methods (KCL in the nodal working on set 2 p.11; KVL in every divider). This file states them on their own, as the syllabus asks.

Build

1 · KCL: what goes into a node comes out

PREDICT

5 mA and 3 mA flow into a node. 2 mA flows out along a third wire. What flows in the fourth wire, and which way?



One node, four wires.

Kirchhoff's current law (KCL): the algebraic sum of the currents at any node is zero. Equivalently, **total current in = total current out**.

$$\sum i_{\text{in}} = \sum i_{\text{out}}$$

Why it holds: charge cannot pile up at a node (a node stores no charge). So every coulomb that arrives per second must leave.

For the Predict: $5 + 3 = 2 + i_x$, so **$i_x = 6 \text{ mA}$, out**. If you had drawn i_x pointing in, you would get $i_x = -6 \text{ mA}$: the same answer. A minus sign only means the real current flows against your arrow.

You already used KCL in 02: the three branch currents of the class p.5 question add back to the source, $5.33 + 2.67 + 4 = 12 \text{ A}$.

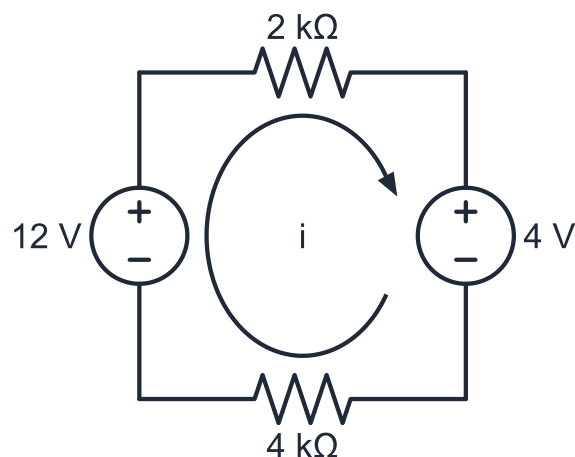
CHECK 1

Currents into a node: 4 mA , -1 mA , i . Out of it: 7 mA . Find i .

2 · KVL: round a loop, the voltages add to zero

TEXTBOOK

Find the loop current i , the voltage across each resistor, and the power of every element.



The 4 V source has $+$ at the top, so it pushes against the 12 V source.

GUESS FIRST

Which source "wins", and is the current in the direction of the arrow shown?

Kirchhoff's voltage law (KVL): the algebraic sum of the voltages around any closed loop is zero.

$$\sum v_{\text{rises}} = \sum v_{\text{drops}} \quad (\text{round any closed loop})$$

Why it holds: voltage is energy per unit charge. A charge that travels round a loop and returns to its start has the same energy it began with.

Method. Go round the loop in the direction of i (clockwise). A source traversed from $-$ to $+$ is a **rise**. A resistor traversed along i is a **drop** of iR . A source traversed from $+$ to $-$ is a **drop**.

- 12 V source, bottom to top (– to +): rise of 12 V.
- 2 kΩ: drop 2i. 4 V source, top to bottom (+ to –): drop 4 V. 4 kΩ: drop 4i.

$$12 = 2i + 4 + 4i \quad \Rightarrow \quad i = \frac{8}{6} = \frac{4}{3} \text{ mA} \approx 1.33 \text{ mA}$$

Voltages: $v_{2k\Omega} = 2 \times \frac{4}{3} = \mathbf{2.67 \text{ V}}$, $v_{4k\Omega} = \mathbf{5.33 \text{ V}}$. Check round the loop: $12 = 2.67 + 4 + 5.33 \checkmark$.

The Guess answer: the 12 V source wins, and $i > 0$ means the current does flow clockwise.

CHECK 2

Swap the 4 V source (+ at the bottom, so it now helps). Find i.

3 • The power check (02 §4, in use)

Sign rule from 02: current **entering** + means the element absorbs $p = vi$.

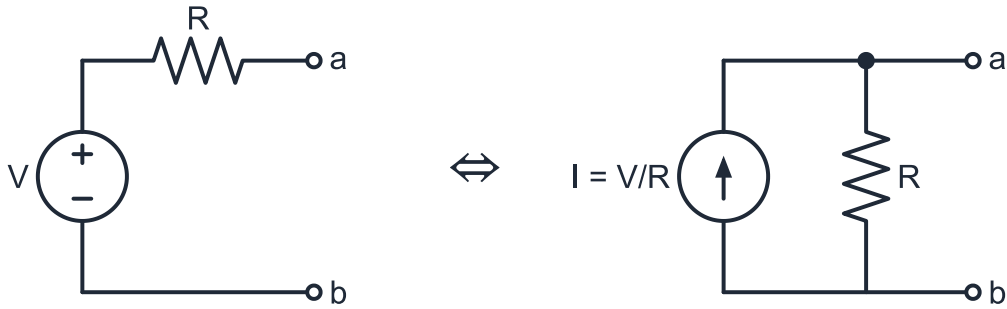
Element	Current enters its...	Absorbed power
12 V source	– terminal (current leaves +)	$-12 \times \frac{4}{3} = \mathbf{-16 \text{ mW}}$ (delivers 16 mW)
4 V source	+ terminal	$+4 \times \frac{4}{3} = \mathbf{+5.33 \text{ mW}}$ (it is being charged)
2 kΩ	+ (always, for a resistor)	$(\frac{4}{3})^2 \times 2 = \mathbf{+3.56 \text{ mW}}$
4 kΩ	+	$(\frac{4}{3})^2 \times 4 = \mathbf{+7.11 \text{ mW}}$
Sum		$-16 + 5.33 + 3.56 + 7.11 = \mathbf{0 \checkmark}$

Delivered = absorbed. Use this as the last line of any loop or node answer: if the powers do not add to zero, something upstream is wrong.

4 • Source transformation

PREDICT

A 10 V source with 2 kΩ in series. Short its terminals: what current flows? Leave them open: what voltage appears? Could a current source with a resistor in parallel give the same two answers?



Both give the same open-circuit voltage (V) and the same short-circuit current (V/R), so no outside circuit can tell them apart.

A **voltage source V in series with R** is equivalent to a **current source $I = V/R$ in parallel with the same R** (class p.4; set 2 p.4 writes $I = V/R$ under the current source), as seen from the terminals a–b.

For the Predict: short-circuit current = $10/2k = \mathbf{5\text{ mA}}$; open-circuit voltage = $\mathbf{10\text{ V}}$. So it is equivalent to $5\text{ mA} \parallel 2\text{ k}\Omega$, and $5\text{ mA} \times 2\text{ k}\Omega = 10\text{ V}$, the same terminal behaviour.

It is the bridge between Thévenin and Norton in 06: $I_N = V_{Th} / R_{Th}$.

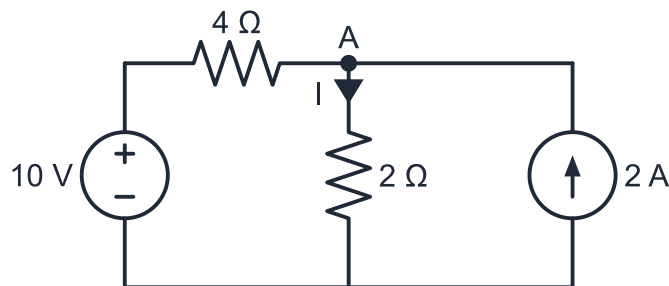
ONLY WORKS WITH A RESISTOR

An **ideal** voltage source (no series R) has no current-source equivalent: $V/0$ is undefined. Same for an ideal current source with no parallel R .

5 · Using it on a class circuit

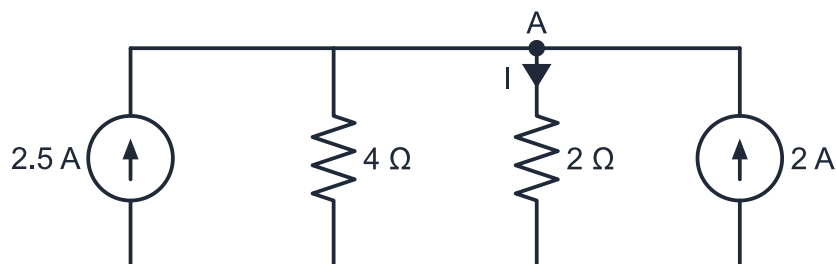
CLASS P.7 · SET 2 P.8

Find I in the superposition circuit (10 V , $4\text{ }\Omega$, $2\text{ }\Omega$, 2 A) without superposition.



The class circuit (05 §1).

Transform 10 V in series with $4\text{ }\Omega$ into $\frac{10}{4} = \mathbf{2.5\text{ A}}$ in parallel with $4\text{ }\Omega$:



Now everything is in parallel between node A and the bottom rail.

Both current sources push up into A: $2.5 + 2 = 4.5$ A. It flows into $4\ \Omega \parallel 2\ \Omega = \frac{4}{3}\ \Omega$, so $V_A = 4.5 \times \frac{4}{3} = \mathbf{6\ V}$ and $\mathbf{I = \frac{6}{2} = 3\ A}$, the professor's answer (05 §1) in three lines.

CHECK 3

Convert 3 mA in parallel with 4 k Ω into a voltage-source form.

Exam form

- **KCL:** Σ currents into a node = Σ currents out (charge is conserved; a node stores no charge).
- **KVL:** Σ voltage rises = Σ voltage drops round any closed loop (energy is conserved).
- **Sign rule:** a negative answer means the real direction is opposite to the arrow you drew. Keep your arrow; report the sign.
- **Power balance:** Σ absorbed power = 0 over the whole circuit.
- **Source transformation:** V in series with R $\Leftrightarrow$ V/R in parallel with R. Not for ideal sources.

Traps

- Changing sign conventions half-way round a loop. Fix the direction once and apply the rise/drop rule to every element.
- Source transformation keeps R **the same size**. It changes from series to parallel, not in value.
- Transforming a source whose resistor is the one you are asked about: that resistor disappears into the equivalent. Keep the element you need outside the transformation.

Self-test

1. State KCL and KVL, each with the conservation law behind it.
2. A node has 10 mA in and three outputs: 2 mA, 5 mA, i. Find i.
3. A loop has 9 V, 1 k Ω and 2 k Ω . Find i and check the power balance.
4. Transform 12 V in series with 3 k Ω .
5. Why is a negative answer from KCL not a mistake?

Answers

Check 1. $4 + (-1) + i = 7 \Rightarrow i = 4 \text{ mA}$ into the node.

Check 2. $12 + 4 = 6i \Rightarrow i = 8/3 \approx 2.67 \text{ mA}$.

Check 3. $3 \text{ mA} \times 4 \text{ k}\Omega = 12 \text{ V}$ in series with $4 \text{ k}\Omega$.

Self-test.

1. KCL: $\sum i_{\text{in}} = \sum i_{\text{out}}$ at a node (conservation of charge). KVL: $\sum v$ round a closed loop = 0 (conservation of energy).
2. $10 = 2 + 5 + i \Rightarrow i = 3 \text{ mA}$.
3. $i = 9/3\text{k} = 3 \text{ mA}$. Source delivers $9 \times 3 = 27 \text{ mW}$; resistors absorb $9 + 18 = 27 \text{ mW}$ ✓.
4. **4 mA in parallel with 3 k Ω .**
5. It only means the current flows opposite to the arrow you chose; the magnitude is still right.